

LabRack Vision QA

YOLO11s/960 + OpenCV · staged, non-patient images · human review required

Kate and I support rack checks without replacing judgment



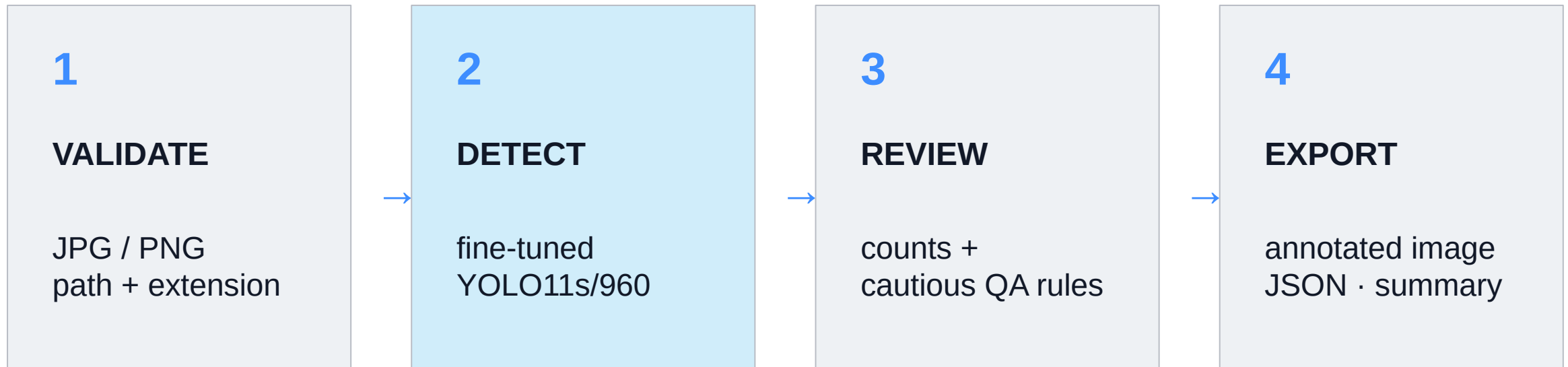
Our prototype

We detect racks, caps, and possible empty slots in a still image.

Our boundary

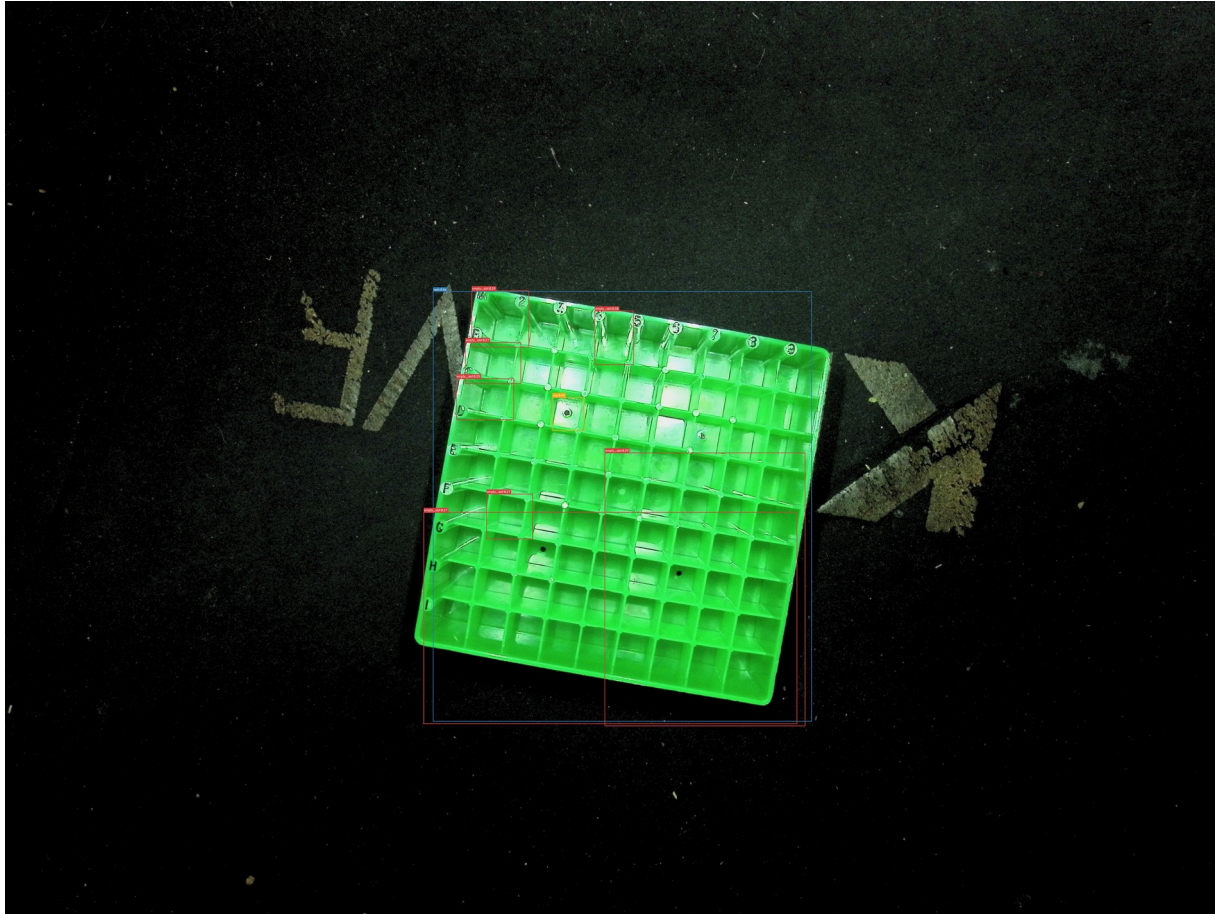
We flag possible issues for human review. We use no patient data and make no clinical claims.

What we built: one image becomes three artifacts



Our fine-tuned detector feeds cautious QA rules and three reviewable outputs.

Our demo produces image, JSON, and review output



CLI SUMMARY

rack	1
tube	0
cap	1
empty_slot	7

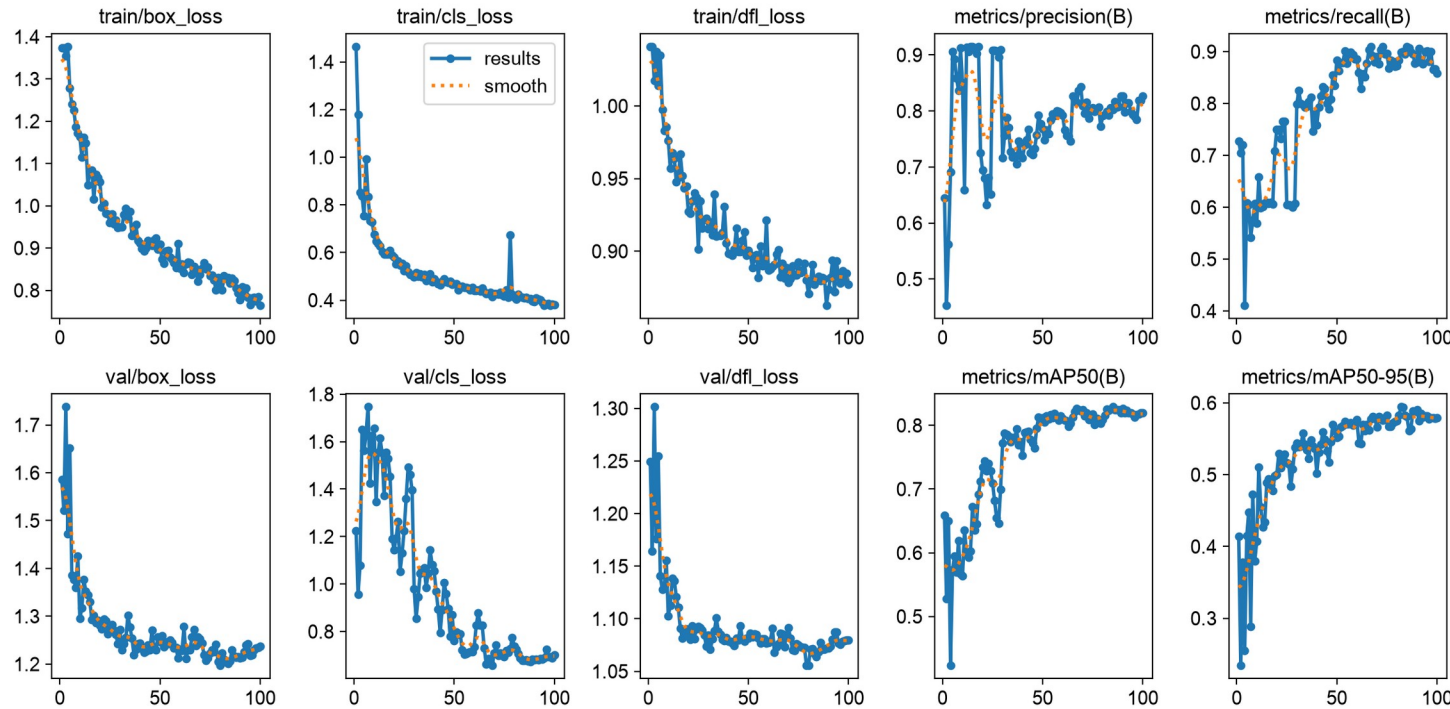
8 detections below 0.50
→ **manual review recommended**

Detection step: 0.732 s

How we built it: YOLO11s transfer learning at 960 px

RUN

100 epochs
960 px · batch 2
YOLO11s



HARDWARE

Apple M3 Pro
PyTorch MPS

We observed MPS
non-deterministic-operation warnings.

We used a grouped split to keep rack images apart

140

staged images
partial Roboflow import

125

TRAIN · Rack_A/B

5

VALIDATE · Rack_C

10

TEST · Rack_D

Current annotation inventory

rack 133 cap 7,391 empty_slot 3,339

Kate and I built this project together. Roderick handled programming + debugging.

<https://www.linkedin.com/in/rodjtaylor/>

Kate contributed photos + annotations and helped us design the testing methodology.

<https://www.linkedin.com/in/kate-leemann-a21615236/>

Our results versus our Blueprint targets

0.937

target ≥ 0.75
test mAP50

2.07 s

target < 3 s
full CLI wall time

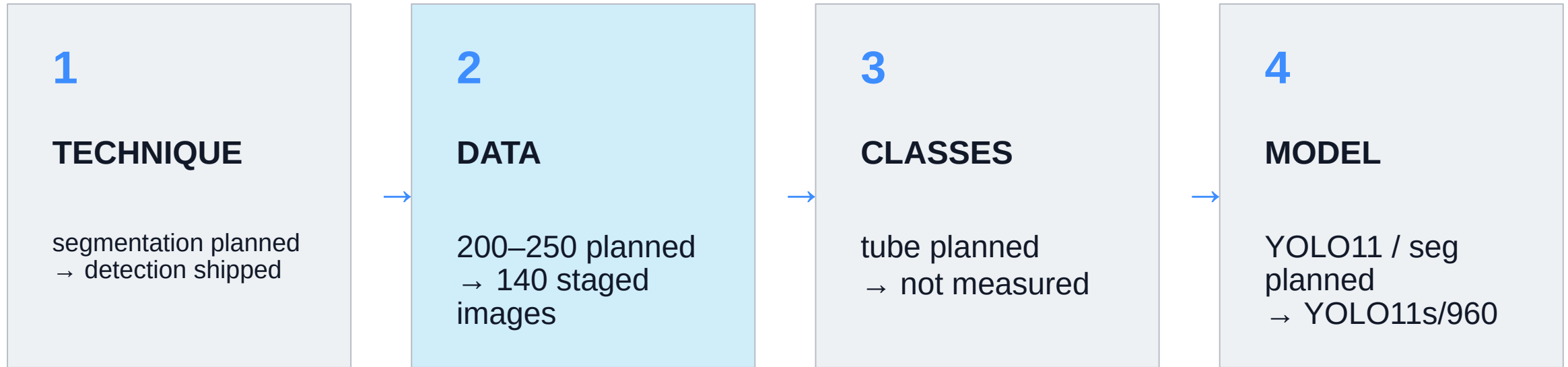
3 outputs

target: annotated
image + JSON + summary

Rack_D test: 10 staged images · 822 instances · mAP50–95 0.601

We met both numeric targets; our failure cases still require human review.

What we changed from our Blueprint



We changed the scope to fit our staged data; we kept the end-to-end promise.

We fixed three issues to get our prototype working

960 px

small empty slots
needed more detail

SPLIT

rack-level split
reduced leakage

BATCH 2

MPS smoke tests
proved the model fit

We changed resolution, grouping, and memory settings one at a time.

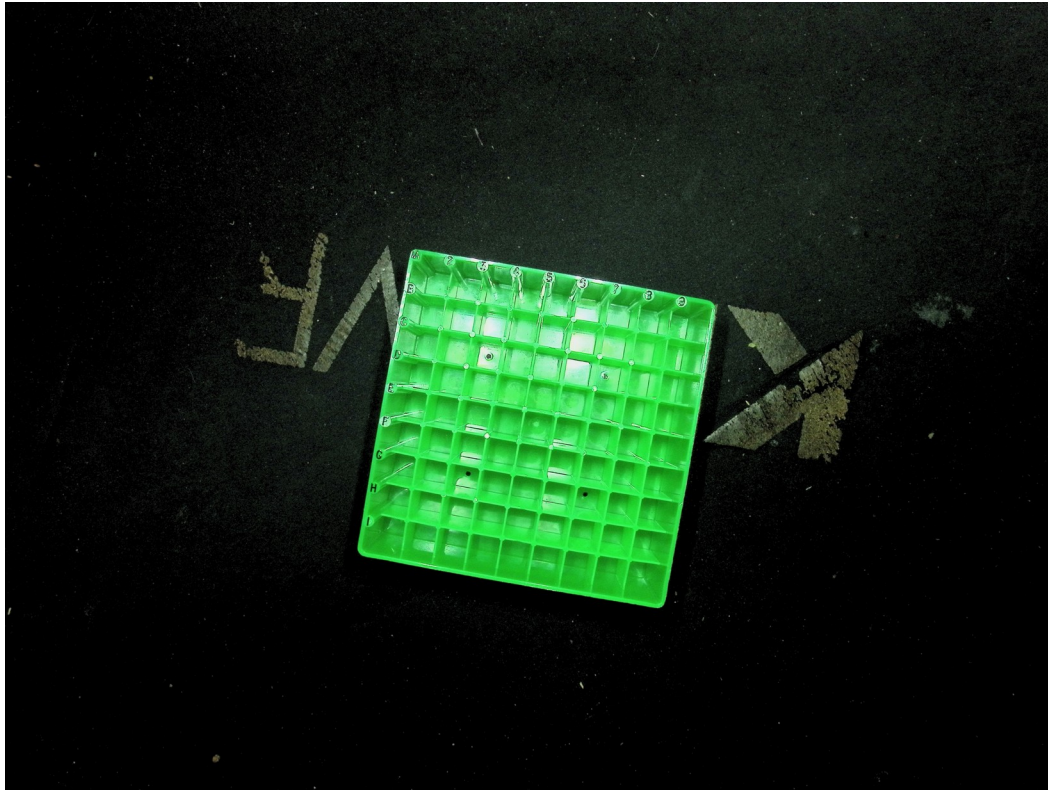
We expect slight MPS variation; every QA finding still requires human review.

Our pipeline works; our dataset must grow.

- 1 We protect evaluation with grouped splits
- 2 We use 960 px for small slots
- 3 We need clearer staged tube images
- 4 We will add a new rack group + labels

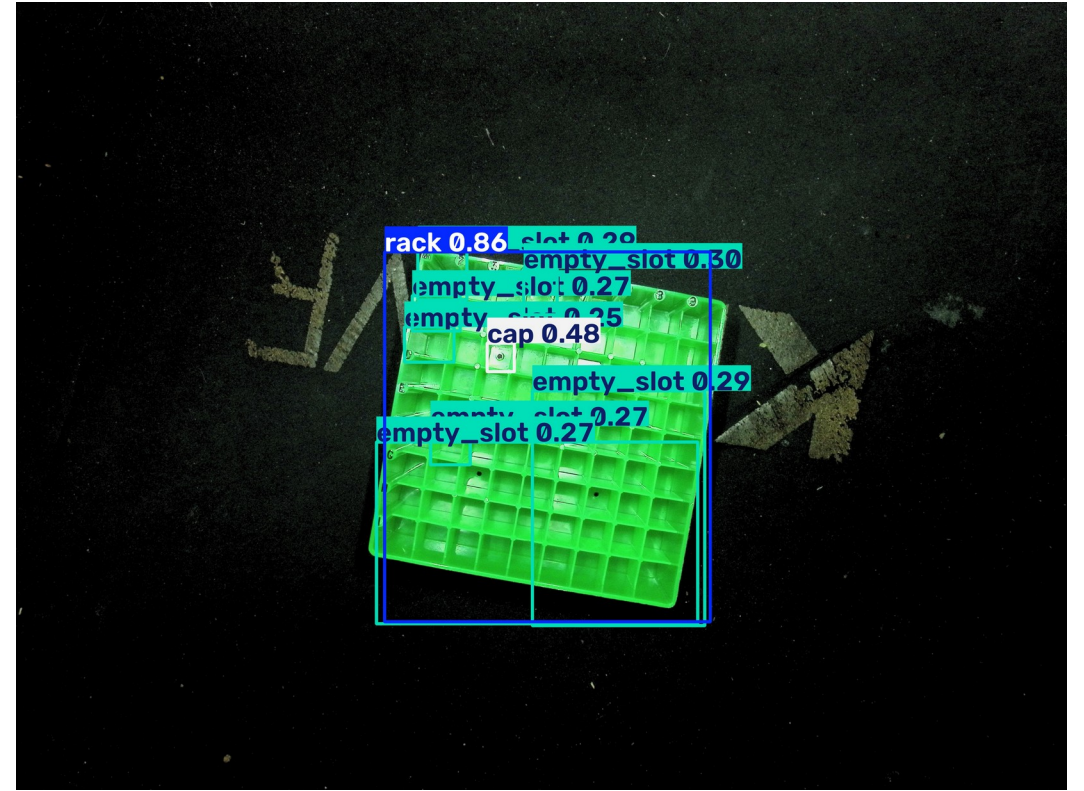
Human review required.
[github.com/RTaylor89/](https://github.com/RTaylor89/labrack-vision-qa)
[labrack-vision-qa](https://github.com/RTaylor89/labrack-vision-qa)

Our failure case: 7 of 81 empty slots detected



KATE'S ANNOTATION

81 empty · 0 caps



OUR MODEL OUTPUT

7 empty · 1 cap